

KARTHIK VAKKALAGADDA

Software Engineer in Test | Test Infrastructure, CI/CD & AI-Assisted Engineering

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SUMMARY

- 6+ years building test automation frameworks and CI/CD test infrastructure for high-volume digital ordering, POS, and payments platforms across four Inspire Brands properties — Web, Mobile, and API — orchestrated through GitLab CI with BrowserStack-backed parallel execution and quality gates.
- Framework owner from year one: authored the majority of the Java automation framework for Arby's and Buffalo Wild Wings, grew those suites to 2,000+ automated tests, reviewed and merged 200+ merge requests across a shared multi-brand codebase, and now serve as technical lead for a 4–6 person distributed US/India team.
- Sole architect and developer of a unified QA platform (personal project) built on MCP servers, agent skills, and prompt-driven test generation, with working GitLab CI and Jira/Xray integrations; sanitized rebuild published on GitHub (shuriken, play-left).

CORE COMPETENCIES

- CI/CD and developer infrastructure: GitLab pipeline design, scheduled/on-demand execution, configurable brand/environment/suite selection, quality gates, parallel and cross-device execution on BrowserStack
- Build and dependency management for automation codebases: Gradle, config registry and loader design, repository restructuring, secrets management, security hardening
- Integration and system testing of distributed services: REST APIs, Azure Service Bus message flows, service-layer migrations, data-integrity validation in SQL and MongoDB
- Automation framework architecture (Web, Mobile, API) with shared runners, reusable flow libraries, configurable multi-brand/multi-environment execution
- Reliability and resiliency validation: circuit breakers, induced-failure and fault-tolerance scenarios, release-readiness quality gates and reporting
- Product quality practice: POS and payments integration testing (Micros, Infor, NCR, BrinkPOS), POS-to-platform reconciliation, A/B and feature-flag validation (Optimizely, Contentful), accessibility testing (UsableNet, VoiceOver, TalkBack), UAT
- AI-assisted engineering workflows: MCP servers, agent skills, tool-use orchestration, prompt-driven generation with review gates, LLM-based evaluation
- Technical leadership: design/standards ownership, merge-request review at volume, mentoring across onshore/offshore teams

TECHNICAL SKILLS

- **Languages:** Java (primary), Python, TypeScript/JavaScript, SQL, Kotlin (personal projects)
- **Cloud & Messaging:** AWS (Cloud Practitioner certified), Azure Service Bus and Azure-hosted databases (integration validation), Docker, Vercel, Cloudflare Pages
- **CI/CD & Build Tooling:** GitLab CI/CD, GitHub Actions, Gradle, Vite, Git, BrowserStack device cloud, secrets management, test orchestration, parallel execution
- **Automation & Frameworks:** Selenium WebDriver, Appium, Playwright, Maestro (incl. Maestro AOS TE), REST Assured, TestNG, Cucumber BDD (Gherkin), Tosca
- **AI Tooling:** MCP servers, agent skills, Claude, GitHub Copilot, prompt-driven test generation, LLM-based evaluation frameworks
- **Data, APIs & Delivery:** SQL, MongoDB, REST APIs, Postman, Swagger, Jira, Xray, qTest, Confluence, ReportPortal, Figma, Contentful

PROFESSIONAL EXPERIENCE

SDET / Automation Technical Lead — Multi-Brand Digital Ordering & POS Platforms Apr 2026 – Present

Capgemini (Client: Inspire Brands) | Atlanta, GA

- Own automation standards, framework direction, and CI/CD test execution across four brands (Sonic, Buffalo Wild Wings, Dunkin', selected Arby's work) for their digital ordering and POS service platforms; triage pipeline and environment failures to root cause and drive systemic fixes.
- Built Sonic E2E regression coverage using AI agent skills and prompt-driven test generation, establishing a standardized author-review workflow with human validation gates for AI-generated tests across the team.
- Dunkin' loyalty platform: resiliency and fault-tolerance testing under induced failure conditions, including circuit-breaker validation across service dependencies.
- Buffalo Wild Wings Business Service Layer (BSL) migration: validated API contracts, regression, and cross-service integration paths through the service-layer change.
- Technical lead for a 4–6 person US-onsite / India-offshore team — defining framework standards, test infrastructure, and reporting conventions, mentoring engineers, and driving adoption of shared tooling.

SDET / Automation Lead — Sonic E2E Digital Ordering & POS Platform Jan 2024 – Mar 2026

Capgemini (Client: Inspire Brands) | Atlanta, GA

- Owned test strategy and automation architecture for Sonic’s end-to-end digital ordering and POS integration platform across Web, Mobile, and API layers.
- Architected the CI/CD test infrastructure: GitLab pipelines with BrowserStack execution, quality gates, and Slack/email reporting for parallel runs and fast feedback — improving release visibility and removing manual release prep.
- Designed and built the Maestro mobile E2E framework (shared flows, smoke suites, critical-path coverage); introduced standardized runners (Playwright for web, Maestro AOS TE for Android) reused across brands.
- Drove repository and domain restructuring — config registry, loader updates, security hardening — and standardized deployment flows across automation and menu domains.
- Validated A/B experiments and feature-flag rollouts (Optimizely, Contentful) and performed accessibility testing (UsableNet, VoiceOver, TalkBack) across mobile and web surfaces.

SDET — Sonic POS, Payments & Digital Integration May 2023 – Dec 2023

Capgemini (Client: Inspire Brands) | Atlanta, GA

- Built and maintained automated regression and smoke suites validating integrations between digital ordering services and POS systems (Micros, Infor) across ordering, payment, and pricing workflows.
- Backend API testing with REST Assured and data-integrity validation in SQL and MongoDB; reconciled POS records against ordering-platform and payment-processor data to catch pricing and payment mismatches.

SDET — Buffalo Wild Wings (BWW Choice) May 2022 – Apr 2023

Capgemini (Client: Inspire Brands) | Bengaluru, India

- Sole automation engineer for BWW Choice mobile web and API layers — built the in-store dining automation suites in Java (Selenium, REST Assured), integrated into GitLab CI/CD with ReportPortal and Jira reporting.

Automation Engineer, Web & API Lead — Arby’s & Buffalo Wild Wings Jul 2020 – Apr 2022

Capgemini (Client: Inspire Brands) | Bengaluru, India

- Authored the majority of the Java automation framework (Selenium, Appium, REST Assured, Cucumber BDD, TestNG) from its early stages — flow structure, reusable utilities, test data handling, and reporting.
- Owned Web and API automation end to end as sole engineer on both layers and co-owned mobile app automation, growing the Arby’s and BWW suites to 2,000+ automated tests.
- Reviewed and merged 200+ merge requests across a shared multi-brand codebase and configured GitLab CI/CD pipelines for scheduled and on-demand execution with configurable brand/environment/suite selection.

Test Automation Engineer — T-Mobile Customer Care & Retail Jan 2020 – Jun 2020

Capgemini (Client: T-Mobile) | Bengaluru, India

- Tested desktop, web, mobile, and API applications using Tosca and Appium (Java); validated APIs with Postman and tracked defects in Jira.

ENGINEERING PROJECTS & PUBLIC PORTFOLIO

- **Unified QA Platform (personal project)** — Designed and built the platform architecture as sole developer: test orchestration, CI/CD tooling, Jira/Xray integration, web dashboard, and supporting utilities (Gherkin-to-Jira importer, health checks, Xray coverage dashboards) across Sonic, Arby’s, and Buffalo Wild Wings.
- **AI-Assisted Test Generation** — Built MCP servers, agent skill libraries, prompt-driven generation runners, and EPIC/TE test-case creation runners on top of the platform, with structured handling of model outputs in the authoring workflow.
- **shuriken (GitHub)** — Sanitized public rebuild of the platform’s core automation architecture with dummy data, including its MCP servers, agent skills, and prompt-driven runners.
- **play-left (GitHub)** — Standalone multi-brand Playwright web E2E framework: brand facts in a typed registry instead of forked specs, plus a dependency-free demo storefront so CI needs no external network.
- **Personal Tracker (live)** — Calendar-centric tracker for concurrent goals; TypeScript/Vite, local-first IndexedDB, tested with Playwright and Vitest, Dockerized, GitHub Actions CI.
- **Cricket Field Planner (live)** — Android (Kotlin) and web (React/TypeScript) tool that validates fielder placements against ICC regulations and exports a tactical plan; deployed on Cloudflare Pages, Gemini API integration.

CERTIFICATIONS & EDUCATION

- **Certifications:** AWS Certified Cloud Practitioner • AWS Generative AI Essentials • Tricentis Tosca L1, TQL, qTest & RPA Specialist • Harvard ManageMentor
- **Education:** B.Tech, Electronics & Communication Engineering — IIITDM Jabalpur, India, 2019